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Purchasing-Issuing OPD Tickets in a Government Hospital: A Case Study on Queueing Process Analysis

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1 Introduction :

Queueing theory is a branch of applied probability and statistics that deals with the mathematical study of waiting lines or queues. It helps analyze systems in which customers or patients arrive for service, wait in a queue if the service facility is busy, and depart after receiving service. Queueing models are widely used in healthcare, banking, transportation, telecommunication, and manufacturing systems to evaluate performance measures such as waiting time, queue length, server utilization, and system efficiency.

In healthcare institutions, especially in Outpatient Department (OPD) service counters, efficient patient management is extremely important because large patient arrivals and limited service facilities often lead to congestion and long waiting times. Excessive waiting not only affects patient satisfaction but also reduces the operational efficiency of the hospital system. Therefore, statistical and queueing analysis of OPD services can help identify bottlenecks and improve resource allocation and service quality.

The present project focuses on the analysis of the OPD counter service system at Barasat Government Medical College and Hospital. Real-life arrival and service data were collected from the OPD counters of the hospital to study the behavior of the queueing process. The primary objective of this study is to model the patient arrival and service mechanisms using suitable probabilistic and statistical methods and to determine an appropriate queueing model for the system.

In this project, special attention is given to the stochastic behavior of inter-arrival and service times. The study examines whether classical assumptions such as Poisson arrivals and exponential service times are valid for the observed data. Since healthcare systems often show time-dependent and irregular arrival patterns, alternative distributions and stochastic processes are also explored. Statistical tools such as exploratory data analysis, graphical visualization, goodness-of-fit tests, and distribution fitting techniques are used to analyze the collected data.

The project further investigates important queueing characteristics including arrival rate, service rate, server utilization, traffic intensity, and waiting behavior of patients in the OPD counter system. The analysis is performed using statistical software and computational methods to provide both theoretical and practical insights into the functioning of the hospital service process.

The findings of this study may help improve patient flow management and reduce congestion in OPD services by providing a better understanding of the queue dynamics. Overall, this project demonstrates the practical application of queueing theory and statistical modeling in analyzing and improving real-world healthcare service systems.

2 Data Collection Overview :

To empirically validate our queueing model and estimate its operational parameters, primary data was collected through direct continuous observation at the Outpatient Department (OPD) ticket counters of **Barasat Government Medical College and Hospital**. The fieldwork was structured to capture the real-time stochastic nature of patient arrivals, counter service times, and physical line congestion under state-dependent conditions.

The dataset captures the dynamics of a multi-server framework operating in a high-volume medical facility. The empirical observation framework is broken down into three core components:

- **Patient Arrivals:** A total of **690 observations** of precise patient arrival times were recorded. These timestamps were utilized to calculate consecutive inter-arrival times, establishing the basis for analyzing the arrival counting process and computing the operational arrival rate (λ_n) across distinct time intervals.
- **Service Completion:** A total of **263 observations** tracking individual patient service entry times (“Service Time IN”) and service completion times (“Service Time OUT”) were logged. The resulting inter-service durations allow for direct mathematical calculation of the state-dependent service rate (μ_n).
- **Queue Dynamics:** The physical queue lengths across two operational counters (Queue 1 and Queue 2), along with the system length (n), were documented at fixed time intervals to monitor system congestion levels and fluctuating line lengths.

2.1 Sample Representation of Collected Dataset :

Below are representative subsets of the primary empirical data collected during the hospital study, demonstrating the exact structural layout of the observed arrival logs, service times, and multi-queue state tracking variables.

2.2 Patient Arrival and Inter-Arrival Time Dataset Sample :

This subset displays the precise tracking of the incoming patient stream and the calculated time intervals (in minutes) between consecutive arrivals.

No. of Patients	Arrival Time	Inter Arrival Time (in min.)
1	8:37:28	0
2	8:37:49	0.35
3	8:38:23	0.566666667
4	8:40:11	1.8
5	8:40:25	0.233333333
6	8:40:50	0.416666667
7	8:41:10	0.333333333
8	8:41:25	0.25
9	8:42:27	1.033333333
10	8:42:40	0.216666667
11	8:42:55	0.25
12	8:43:12	0.283333333
13	8:43:20	0.133333333
14	8:43:28	0.133333333
15	8:43:36	0.133333333

Table 1: Sample Dataset of patient arrivals and calculated inter-arrival times.

2.3 Patient Service and Inter-Service Time Dataset Sample :

This segment details the exact tracking when a patient began service versus when they cleared the counter window, detailing the absolute processing time.

No. of Patients	Service Time (IN)	Service Time (OUT)	Inter-Service Time (In Min.)
1	8:35:00	8:38:00	3
2	8:36:00	8:39:00	3
3	8:38:00	8:39:00	1
4	8:38:00	8:40:00	2
5	8:39:00	8:41:00	2
6	8:39:00	8:42:00	3
7	8:40:00	8:42:00	2
8	8:40:00	8:42:00	2
9	8:42:00	8:44:00	2
10	8:42:00	8:43:00	1
11	8:43:00	8:44:00	1
12	8:42:00	8:44:00	2
13	8:44:00	8:45:00	1
14	8:44:00	8:45:00	1
15	8:45:00	8:46:00	1

Table 2: Sample dataset patient counter service durations.

2.4 State-Space and Multi-Queue Length Status Sample :

This subset captures system snapshots across all two individual ticket counter lines, mapping the collective system size metrics at specific designated checkpoints.

Time	Queue 1	Queue 2	System Length(n)	Total Queue Length
8:35	1	3	10	4
8:45	1	3	15	4
8:50	3	11	14	14
9:00	3	5	25	8
9:10	6	9	30	15
9:20	10	10	40	20
9:25	8	10	24	18
9:30	8	16	28	24
9:35	13	27	45	40
9:40	14	33	60	47

Table 3: Dataset of system tracking state parameters and line accumulation points.

3 Description of Poisson Process :

In queueing theory, the **Poisson process** serves as the foundational mathematical framework for modeling how customers arrive at a service system. It is a specific type of continuous-time Markov chain that counts the number of events (arrivals) that occur over a given time interval.

A Poisson process assumes that arrivals occur randomly and completely independently of one another. In a standard queueing model (such as the classical $M/M/1$ queue model), the process is characterized by three core properties:

- **Independent Increments:** The number of arrivals occurring in two non-overlapping time intervals are completely independent. Past arrival patterns do not influence future arrivals.
- **Stationary Increments:** The probability of a certain number of arrivals happening in a time interval depends on the *length* of the interval, not on its specific starting time.
- **Rare Simultaneous Events:** The probability of multiple arrivals occurring at the exact same split second (an infinitesimally small time interval Δt) is asymptotically zero.

Since the time between successive arrivals in a Poisson process is **exponentially distributed**, it possesses the **memoryless property**. Therefore, given that no arrival has occurred for a certain period, the distribution of the remaining waiting time until the next arrival remains unchanged.

4 Notations and System Variables :

To formally define the states and underlying dynamics of the queueing system over time, we introduce the following stochastic processes and state-dependent parameters:

4.1 Number of Arrivals $X(t)$:

$X(t)$ is the counting process that tracks the **number of arrivals in the system (arrivals)** that have occurred from the beginning of observation (time 0) up to time t .

4.2 Number of Departures $Y(t)$:

$Y(t)$ is the counting process that tracks the **number of service completions (departures)** that have occurred from the beginning of observation (time 0) up to time t .

4.3 Number of Services $N(t)$:

$N(t)$ is the number of services takers in the system (here in hospital) at time t .

4.4 Arrival Rate $\lambda_n(t)$:

$\lambda_n(t)$ denotes the **arrival rate** (expected number of arrivals per unit time) given that the system is exactly in state n (i.e., $X(t) = n$) at time t .

- In a basic queueing system, this is constant: $\lambda_n(t) = \lambda$.
- Modeling it as state-dependent allows the framework to capture complex real-world behaviors such as *balking*, where incoming customers refuse to join if the queue length n is already too large.

4.5 Service Rate $\mu_n(t)$:

$\mu_n(t)$ signifies the **service rate** (expected number of departures per unit time) when the system holds n customers at time t .

- In a single-server system ($M/M/1$), the service rate remains constant for any active system: $\mu_n(t) = \mu$ for all $n \geq 1$.
- In a multi-server setup ($M/M/c$), the service rate scales directly with the number of busy servers, yielding:

$$\mu_n(t) = \begin{cases} n\mu & \text{if } 0 \leq n < c \\ c\mu & \text{if } n \geq c \end{cases} \quad (1)$$

5 Assumptions of Poisson Process :

For a counting process to be classified as a Poisson process with a constant arrival rate $\lambda > 0$, it must satisfy strict probabilistic conditions regarding how events occur over infinitesimally small increments of time, denoted as Δt .

These foundational assumptions ensure that the underlying process remains entirely random, memoryless, and devoid of joint or simultaneous occurrences (such as two distinct customers entering a service channel at the exact same split second).

5.1 Notations and Mathematical Formulations :

Let $X(t)$ be the counting process representing the total number of arrivals up to time t . Therefore, the term $X(t + \Delta t) - X(t)$ represents the exact number of events or arrivals that occur within the localized small time interval $(t, t + \Delta t]$.

As the time increment approaches zero ($\Delta t \rightarrow 0$), the operational behavior of the process is governed by the following structural assumptions:

5.1.1 Probability of a Single Arrival :

The probability that exactly one arrival occurs in an infinitesimally small time interval Δt is directly proportional to the length of that interval, completely independent of the starting epoch t .

$$P(X(t + \Delta t) - X(t) = 1) = \lambda \Delta t + o(\Delta t) \quad (2)$$

Where λ represents the steady-state intensity or arrival rate of the process.

5.1.2 Probability of Multiple Arrivals :

The probability that more than one arrival occurs within the same localized interval Δt is asymptotically negligible. This condition rules out the possibility of simultaneous multiple arrivals in the model.

$$P(X(t + \Delta t) - X(t) \geq 2) = o(\Delta t) \quad (3)$$

Note on the Little-o Notation $o(\Delta t)$: The term $o(\Delta t)$ denotes a function that approaches zero at a rate strictly faster than Δt itself, formally defined as:

$$\lim_{\Delta t \rightarrow 0} \frac{o(\Delta t)}{\Delta t} = 0 \quad (4)$$

5.1.3 Probability of No Arrivals :

By utilizing the fundamental axiom of total probability, the likelihood of experiencing zero arrivals during the interval Δt is derived as the direct complement of the previous

two scenarios:

$$P(X(t + \Delta t) - X(t) = 0) = 1 - \lambda\Delta t + o(\Delta t) \quad (5)$$

5.1.4 Independence and Stationarity :

- **Independent Increments:** For any set of non-overlapping time intervals, the number of arrivals observed in one interval is statistically independent of the counts registered in any other interval.
- **Stationary Increments:** The probability distribution of the number of arrivals within any given time interval depends solely upon the absolute length of the window Δt , and not on its position along the time horizon t .

This is the standard theory of queueing model. But in our case as we are working with a practical data, now we are going to build the queueing model for our case.

6 Relation of Time vs. Arrival Rate:

To analyze how the arrival process changes over time at the Barasat Government Medical College and Hospital OPD ticket counters, the arrival rate $\lambda(t)$ is plotted against specific 5-minute time intervals across the observation window. To visually evaluate this relationship, a Time vs. Arrival Rate Graph was generated. The graph plots the arrival rate per unit time across consecutive 5-minute intervals starting from the early morning opening rush at 8:35–8:40 AM through mid-morning peaks, and continuing until the afternoon cool-down periods ending around 13:05–13:10 PM. The data points explicitly show that patient arrivals do not follow a flat, constant rate; instead, they fluctuate non-linearly, with intense spikes during mid-morning registration hours and a steady decline as the afternoon approaches.

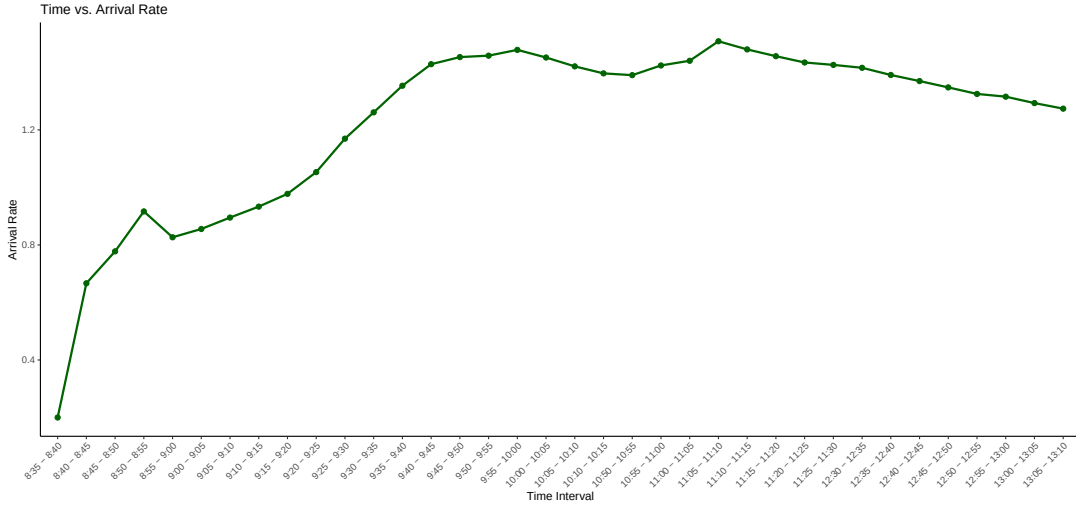


Figure 1: Plot of Time dependent arrival rate vs different time intervals

To approximate this arrival intensity function analytically, we fit a quadratic (degree-2) polynomial regression model to the 36 empirical data points extracted from this timeline. This continuous function allows us to map the acceleration and deceleration of patient traffic smoothly. The parameters of the intensity curve are calculated using the classical method of Ordinary Least Squares (OLS) estimation.

6.1 Least Squares Formulation :

Let t denote the time variable, and $\lambda(t)$ represent the corresponding arrival rate. The 2nd-degree polynomial model modeling the trajectory of traffic intensity is defined as:

$$\lambda(t) = \beta_0 + \beta_1 t + \beta_2 t^2 \quad (6)$$

For a dataset containing $n = 36$ extracted data points, the system of equations can be

compactly structured into a standard general linear model matrix format:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (7)$$

As we know to minimize the residual sum of squared errors ($SSE = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}$), we solve the fundamental **normal equations**:

$$(\mathbf{X}^T \mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T \mathbf{y} \quad (8)$$

Isolating the parameter vector yields the unique ordinary least squares estimator:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (9)$$

—

6.2 Empirical Parameter Solutions :

Based on the empirical processing of the data, the estimated parameters are solved as:

$$\beta_0 = 1.1973677394 \quad (10)$$

$$\beta_1 = 0.0551898023 \quad (11)$$

$$\beta_2 = -0.0002322031 \quad (12)$$

Substituting these numerical coefficients yields the estimated analytical model for our arrival intensity process:

$$\lambda(t) = 1.1973677394 + 0.0551898023 t - 0.0002322031 t^2 \quad (13)$$

—

7 Distribution of Inter-Arrival Time :

Let us assume that throughout the time duration of length t , the inter-arrival time follows approximately an exponential distribution with mean $\frac{1}{\lambda(t)}$.

As a result, we can infer that the corresponding arrival process $X(t)$ is a time-dependent Poisson process, where

$$X(t) \sim \text{Poisson}(m(t))$$

where,

$$m(t) = \int_0^t \lambda(x) dx \quad (14)$$

Ref.- As, The Time Dependent Poisson Process in J.Medhi

A key result from queueing theory states that **if inter-arrival times are independent and identically distributed and follow an Exponential distribution with parameter λ , then the counting process follows a Poisson process with mean λt .**

Reference: J. Medhi, *Stochastic Models in Queueing Theory*, 2nd ed., Academic Press, 2002, p. 150.

Also we have found that over the entire time interval of study out of 3 selected time distributions exponential, gamma, weibull. here exponential fits well.

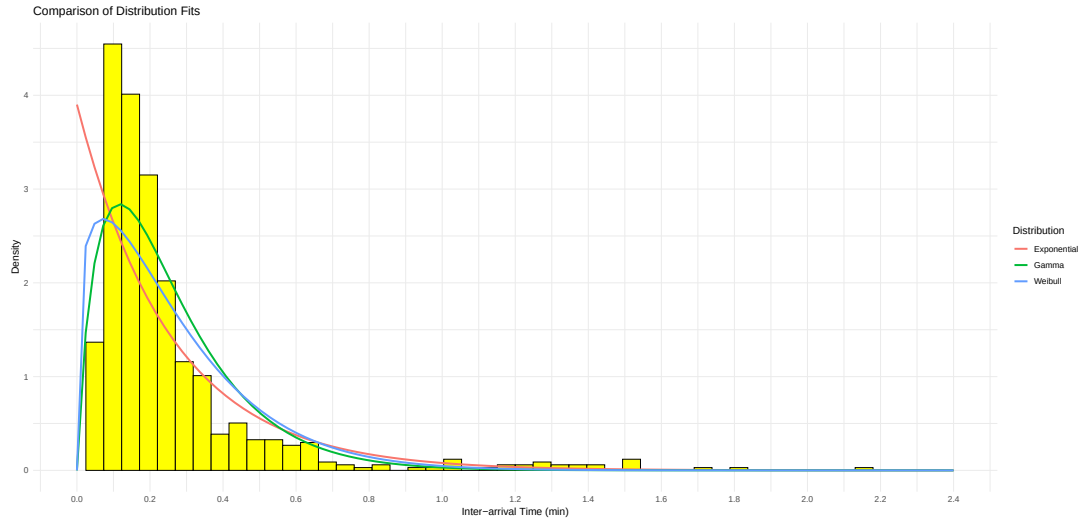


Figure 2: Empirical inter-arrival time density histogram compared against parametric distribution fits.

Now we use AIC and BIC to choose the best model. For both of these, a lower value means a better fit.

Distribution	AIC (Akaike Info Criterion)	BIC (Bayesian Info Criterion)	Rank
Exponential	-683.66	-674.6	1
Gamma	-664.05	-650.45	2
Weibull	-658.12	-644.52	3

Table 4: Goodness-of-fit model indicators for inter-arrival time selections.

We can treat this discussion as a direct or indirect evidence in our assumptions of exponential distribution for our inter arrival time distribution.

8 Time-dependent Poisson Processes (Non-homogeneous Poisson process) :

When modeling traffic over an extended timeline, the classical Poisson process assumption of a constant, homogeneous arrival intensity often fails to align with reality. To accurately capture fluctuations across a day, we transition to a **Time-Dependent Poisson Process** (also known as a **Non-Homogeneous Poisson Process**). Under this framework, the arrival rate is treated as a non-random function of time, denoted by $\lambda(t)$.

The fundamental probabilistic postulates governing an infinitesimally small time increment h (where $h \rightarrow 0$) are formulated as:

- **Probability of exactly 1 arrival ($k = 1$) :**

$$Pr\{X(t+h) - X(t) = 1 \mid X(t) = n\} = \lambda(t)h + o(h) \quad (15)$$

- **Probability of multiple arrivals ($k \geq 2$) :**

$$Pr\{X(t+h) - X(t) = k \mid X(t) = n\} = o(h) \quad (16)$$

- **Probability of zero arrivals ($k = 0$) :**

$$Pr\{X(t+h) - X(t) = 0 \mid X(t) = n\} = 1 - \lambda(t)h + o(h) \quad (17)$$

In this system, the classical postulate regarding homogeneity does not hold good and this process is called a non-homogeneous Poisson process. However, the differential equations of the process can be directly obtained from those of a standard Poisson process by replacing the static parameter λ with the time-varying intensity function $\lambda(t)$:

$$p'_n(t) = -\lambda(t)p_n(t) + \lambda(t)p_{n-1}(t), \quad n \geq 1 \quad (18)$$

$$p'_0(t) = -\lambda(t)p_0(t) \quad (19)$$

To formally analyze this time-dependent counting mechanism, we state and prove the properties of its Probability Generating Function (p.g.f.).

8.1 Theorem for Non-homogeneous Poisson Process

Theorem:

The probability generating function (p.g.f.) of a non-homogeneous Poisson process $\{X(t), t \geq 0\}$ is given by

$$Q(s, t) = \exp\{m(t)(s - 1)\}. \quad (20)$$

where

$$m(t) = \int_0^t \lambda(x) dx \quad (21)$$

is the expected value of $N(t)$.

Source: J. Medhi, *Stochastic Models in Queueing Theory*, 2nd ed., San Diego, CA, USA: Academic Press, 2002, p. 161.

8.2 System State Probabilities :

Corollary 1. The probability of no occurrence in an interval of length t is given by evaluating the generating function at $s = 0$:

$$p_0(t) = Q(s, t)|_{s=0} = \exp\{-m(t)\} = \exp\left\{-\int_0^t \lambda(x) dx\right\} \quad (22)$$

Corollary 2. The probability of exactly k occurrences in an interval $(0, t)$ is extracted via the k -th derivative evaluated at $s = 0$:

$$p_k(t) = \frac{1}{k!} \frac{\partial^k}{\partial s^k} Q(s, t) \Big|_{s=0} = \frac{1}{k!} \{m(t)\}^k e^{-m(t)} \quad (23)$$

8.3 Analytical Derivation of the Cumulative Intensity Function

$m(t)$:

For a non-homogeneous Poisson process, the **cumulative arrival intensity function** (or mean value function) $m(t)$ represents the expected number of arrivals occurring up to time t . It is derived analytically by integrating the continuous arrival intensity function $\lambda(x)$ over the horizon $[0, t]$:

$$m(t) = \int_0^t \lambda(x) dx \quad (24)$$

Substituting our quadratic least squares function into the definite integral:

$$m(t) = \int_0^t (1.1973677394 + 0.0551898023 x - 0.0002322031 x^2) dx \quad (25)$$

Integrating term-by-term with respect to x :

$$m(t) = \frac{1.1973677394}{1}t + \frac{0.0551898023}{2}t^2 - \frac{0.0002322031}{3}t^3 \quad (26)$$

Evaluating the quotients yields the final structural polynomial for the mean value arrival function:

$$m(t) = 1.1973677394t + 0.02759490115t^2 - 0.0000774010333t^3 \quad (27)$$

Rounding the parameters to four significant decimal places for concise implementation, the system function is represented as:

$$m(t) \approx 1.1974t + 0.0276t^2 - 0.000077t^3 \quad (28)$$

This derived mathematical profile for $m(t)$ serves as the structural foundation for proving and evaluating the time-dependent state equations of the queueing framework.

8.3.1 Substitution of Empirical System Metrics :

To customize this theoretical framework for our hospital data collected at the Barasat Government Medical College OPD counters, we substitute the explicit polynomial function $m(t)$ derived from our quadratic least-squares regression:

$$m(t) = 1.1973677394t + 0.02759490115t^2 - 0.0000774010333t^3 \quad (29)$$

Substituting this cumulative intensity polynomial into the general distribution expression yields our final arrival state probability model:

$$p_k(t) = \frac{(1.197368t + 0.027595t^2 - 0.000077t^3)^k \cdot \exp\{-(1.197368t + 0.027595t^2 - 0.000077t^3)\}}{k!} \quad (30)$$

9 Relation of Inter-Service Time and Service Rate:

To define the output channels of our queueing model, we analyze the structural behavior of counter processing speeds. In traditional queueing theory, the continuous time interval separating consecutive patient service completions—defined as the **inter-service time**—is modeled as a continuous random variable.

To determine the mathematical properties of this process, the empirical service rate $\mu(t)$ is mapped against successive 5-minute time intervals across the operational shift.

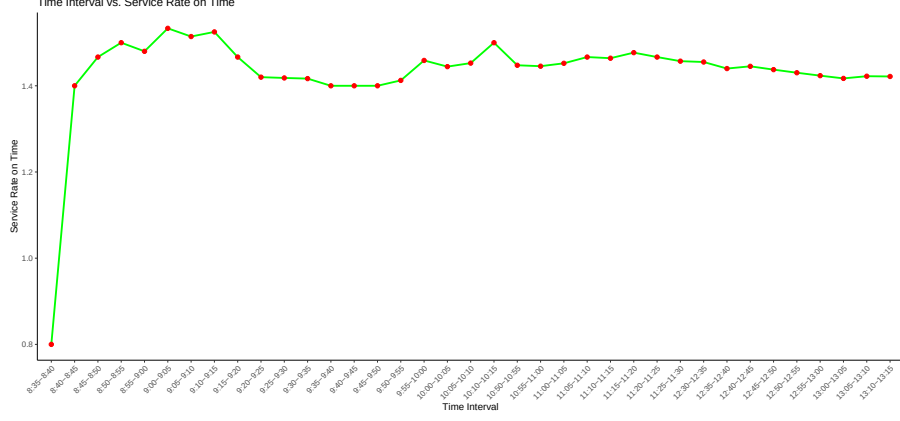


Figure 3: Time Interval vs. Service Rate on Time tracking graph.

By observing the plot we can say that there is no time dependency. Therefore, the complex time-dependent service rate function $\mu_n(t)$ can be simplified directly as:

$$\mu_n(t) = \mu \quad (31)$$

10 Distribution of Inter-Service Times:

To formally capture the stochastic properties of the service channel, five continuous parametric distributions were fitted to the collected inter-service times using R. The models include the Exponential, Gamma, Weibull, Erlang-3, and Erlang-6 distributions.

In our case based on penalized likelihood metrics, the Erlang-6 distribution achieves the minimum absolute information criterion. However, implementing a **higher-order phase distribution introduces extreme computational complications when solving the system's state-space differential equations**. To maintain mathematical tractability while preserving structural accuracy, the **Erlang-3** distribution is selected. In this framework, the Erlang model is treated as a specialized form of the Gamma distribution where the shape parameter is fixed to an integer ($k = 3$).

10.0.1 Distribution Fitting and Comparison:

Figure 4 displays the empirical density histogram of patient inter-service times overlaid with the fitted theoretical density curves.

Although Erlang-6 yields a slightly lower AIC, the Erlang-3 distribution provides a comparable fit while significantly reducing downstream modeling complexity. It allows the service channel to be represented as three sequential, identically distributed exponential phases, creating a robust foundation for building our queueing equations. So it follows the 3-Erlang Renewal Count Process.

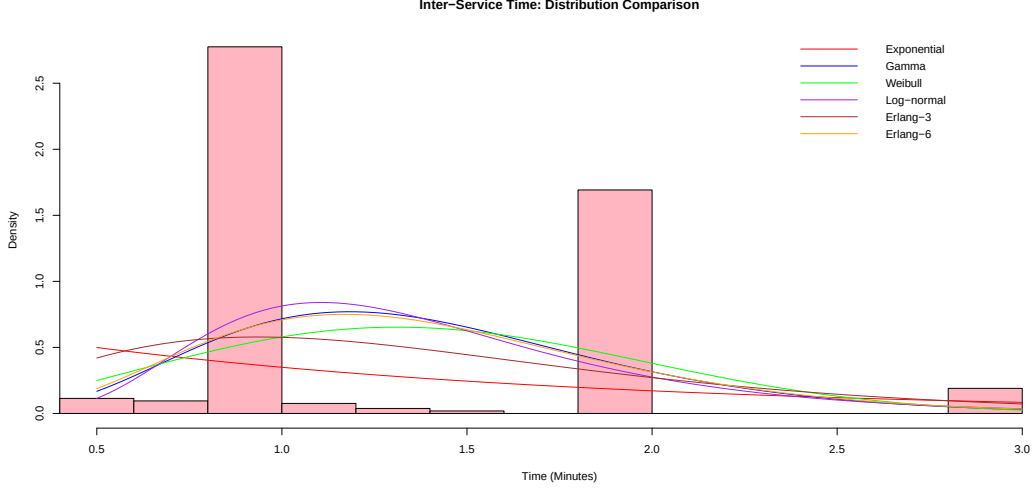


Figure 4: Inter-service time density histogram compared against parametric distribution fits.

10.1 Goodness-of-Fit Criteria Comparison :

To establish the goodness of fit criteria we want to test

The null hypothesis(H_0): the data follows the expected distribution

Alternative hypothesis(H_1): the data does not follow the expected distribution.

Hence in our case the p-value of Erlang-6, Erlang-3, Gamma, Weibull, Exponential distributions all are less than 0.05. So, we reject the null hypothesis. And hence we could not get any information by formulating this test so focus on AIC values.

Table 5 provides the Akaike Information Criterion (AIC) values utilized to evaluate and rank each candidate service distribution.

Rank	Distribution	AIC
1	Erlang-6	410.99
2	Erlang-3	418.0909
3	Gamma	422.49
4	Weibull	442.38
5	Exponential	706.01

Table 5: Goodness-of-fit AIC indicators for patient inter-service times.

11 Distribution of Service Completion Events ($Y(t)$) :

To find the distribution of the service completion process, let the continuous random variable $Y(t)$ denote the total number of completed services within a time interval of length t units. If the individual service completion times follow an Erlang- k distribution,

its probability density function (p.d.f.) is defined as:

$$f(x) = \frac{\mu^k}{\Gamma(k)} e^{-\mu x} x^{k-1}, \quad t > 0; \quad k > 0, \quad \mu > 0 \quad (32)$$

This expression is identical to a Gamma(μ, k) distribution, so an expected service completion time equal to $\frac{k}{\mu}$.

If $S_n = T_1 + T_2 + \dots + T_n$ represent the continuous random variable which is time taken for n service completion. Where, $T_i \sim \text{Gamma}(\mu, k)$ independently, then by the additive property of the Gamma distribution, the sum of these variables is closed under the same scale parameter:

$$S_n \sim \text{Gamma}(\mu, nk) \quad \text{and} \quad S_{n+1} \sim \text{Gamma}(\mu, (n+1)k) \quad (33)$$

The probability that exactly n services are completed during a time interval of length t is,

$$P[Y(t) = n] = P[S_n \leq t] - P[S_{n+1} \leq t] \quad (34)$$

$$\begin{aligned} P[Y(t) = n] &= \int_0^t \frac{\mu^{nk}}{\Gamma(nk)} e^{-\mu u} u^{nk-1} du - \int_0^t \frac{\mu^{(n+1)k}}{\Gamma((n+1)k)} e^{-\mu u} u^{(n+1)k-1} du \\ &= I_{\mu t}(nk) - I_{\mu t}(\overline{n+1}k) \end{aligned} \quad (35)$$

where $I_x(n) = \int_0^x \frac{e^{-u} u^{n-1}}{\Gamma(n)} du$ represents the standard incomplete Gamma integral.

From the general theory of the Poisson distribution, then,

$$P[X \leq n] = 1 - I_\lambda(n+1) \quad (36)$$

Using this identity, the functional difference $I_{\mu t}(nk) - I_{\mu t}(\overline{n+1}k)$ can be rewritten as:

$$I_{\mu t}(nk) - I_{\mu t}(\overline{n+1}k) = \{1 - I_{\mu t}(\overline{n+1}k)\} - \{1 - I_{\mu t}(nk)\} \quad (37)$$

Setting the Poisson parameter boundary to $\lambda = \mu t$,

$$\begin{aligned} &= P[X \leq nk + k - 1] - P[X \leq nk - 1], \quad \text{where } X \sim \text{Poisson}(\mu t) \\ &= P[nk \leq X \leq nk + k - 1] \\ &= \sum_{x=nk}^{nk+k-1} \frac{e^{-\mu t} (\mu t)^x}{x!} \end{aligned} \quad (38)$$

This summation represents the sum of the Poisson probability mass functions evaluated

strictly across the discrete points $nk, nk + 1, \dots, nk + k - 1$.

$$p_n(t) = P[Y(t) = n] = \sum_{x=nk}^{nk+k-1} \frac{e^{-\mu t} (\mu t)^x}{x!}, \quad n = 0, 1, 2, \dots \quad (39)$$

where k is a known positive integer.

Now,

$$E(Y(t)) = \sum_{n=0}^{\infty} n P[Y(t) = n] = \sum_{n=0}^{\infty} n \{P[S_n \leq t] - P[S_{n+1} \leq t]\} \quad (40)$$

$$= \sum_{n=1}^{\infty} n \{P[S_n \leq t] - P[S_{n+1} \leq t]\} \quad (41)$$

$$\begin{aligned} &= (1 \cdot P[S_1 \leq t] - 1 \cdot P[S_2 \leq t]) + (2 \cdot P[S_2 \leq t] - 2 \cdot P[S_3 \leq t]) \\ &\quad + (3 \cdot P[S_3 \leq t] - 3 \cdot P[S_4 \leq t]) + \dots \\ &= P[S_1 \leq t] + P[S_2 \leq t] + P[S_3 \leq t] + \dots \end{aligned} \quad (42)$$

Substituting our incomplete Gamma integral for each term:

$$\begin{aligned} &= I_{\mu t}(k) + I_{\mu t}(2k) + I_{\mu t}(3k) + \dots \\ &= P[X > k - 1] + P[X > 2k - 1] + P[X > 3k - 1] + \dots \end{aligned} \quad (43)$$

Alternatively,

$$E(Y(t)) = \sum_{n=1}^{\infty} n \cdot \sum_{x=nk}^{nk+k-1} \frac{e^{-\mu t} (\mu t)^x}{x!} \quad (44)$$

By changing the order of summation, bounding the inner parameter to $nk \leq x \leq (n+1)k - 1$, which simplifies to limit $n \leq \frac{x}{k}$ and $n \geq \frac{x}{k}$:

$$\begin{aligned} &= e^{-\mu t} \sum_{x=k}^{\infty} \frac{(\mu t)^x}{x!} \sum_{\substack{n \leq [x/k] \\ n \geq [x/k]}} n \\ &= e^{-\mu t} \sum_{x=k}^{\infty} \left[\frac{x}{k} \right] \frac{(\mu t)^x}{x!} \end{aligned} \quad (45)$$

Expanding this expression we have,

$$= e^{-\mu t} \left\{ 1 \cdot \left[\frac{(\mu t)^k}{k!} + \dots + \frac{(\mu t)^{2k-1}}{(2k-1)!} \right] + 2 \cdot \left[\frac{(\mu t)^{2k}}{2k!} + \dots + \frac{(\mu t)^{3k-1}}{(3k-1)!} \right] + \dots \right\} \quad (46)$$

As indicated in standard theory, this expression has no definite, closed-form solution for

general values of k . From different sources it can be approximate it as,

$$\approx \frac{\mu t}{k} \quad (47)$$

To evaluate the probability of exactly one completion over a small time increment $\Delta t \rightarrow 0$, we set $t = \Delta t$ for a single event ($n = 1$):

$$P[Y(\Delta t) = 1] = \sum_{x=k}^{2k-1} \frac{e^{-\mu\Delta t} (\mu\Delta t)^x}{x!} = e^{-\mu\Delta t} \left[\frac{(\mu\Delta t)^k}{k!} + \frac{(\mu\Delta t)^{k+1}}{(k+1)!} + \dots \right] \quad (48)$$

Expanding the exponential term as, $e^{-\mu\Delta t} = 1 - \mu\Delta t + \frac{(\mu\Delta t)^2}{2!} - \dots$:

$$= \left\{ 1 - \mu\Delta t + \frac{(\mu\Delta t)^2}{2!} - \dots \right\} \left[\frac{(\mu\Delta t)^k}{k!} + \frac{(\mu\Delta t)^{k+1}}{(k+1)!} + \dots \right] \quad (49)$$

Since $\Delta t \rightarrow 0$, we drop all higher-order terms ($o(\Delta t)$) beyond the leading power, assuming a small value for the phase parameter k :

$$P[Y(\Delta t) = 1] \approx \frac{(\mu\Delta t)^k}{k!} \quad (50)$$

According to standard queueing theory,

$$P[Y(\Delta t) = 1] \approx \frac{\mu\Delta t}{k} \quad (51)$$

12 Queueing Structure of the OPD Ticketing System

Based on the empirical analysis, patient arrivals are modeled by a non-homogeneous Poisson process with time-varying arrival rate $\lambda(t)$. Let $Y(t)$ denote the number of service completions by time t . The service-completion process $Y(t)$ is assumed to follow an Erlang renewal counting process with probability mass function

$$P\{Y(t) = n\} = e^{-\mu t} \sum_{r=kn}^{k(n+1)-1} \frac{(\mu t)^r}{r!}, \quad n = 0, 1, 2, \dots,$$

where $k = 3$.

Accordingly, the OPD ticketing system is represented by the queueing model

$$M_t/E_3/C/\infty/FCFS.$$

The mathematical formulation of the model is developed in the following section.

13 Definitions and Differential Difference Formulations for the Queueing Model :

We define a multi-server queueing model with c independent operational service channels. Let $N(t)$ denote the random variable representing the total number of patients present in the hospital OPD system at time t .

For the Erlang- k service distribution setup, the effective service completion rate function μ_n when the system occupies state $N(t) = n$ is defined as:

$$\mu_n = n \left(\frac{\mu}{k} \right), \quad 1 \leq n < c \quad (52)$$

$$\mu_n = c \left(\frac{\mu}{k} \right), \quad n \geq c \quad (53)$$

As in our case, the time-dependent arrival intensity $\lambda(t)$ be a concave function of time t having a λ_{max} and λ_{min} within the active observation case. We analyze the system state changes across a small time increment $\Delta t \rightarrow 0$.

13.1 Initial State Probability Derivation ($n = 0$) :

The probability that the system is completely empty at time $t + \Delta t$ requires evaluating the compound probability of starting empty at time t with no new arrivals occurring, or starting with exactly 1 patient in the system who successfully completes all underlying service phases within Δt :

$$P_0(t + \Delta t) = P_0(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} + P_1(t) \left\{ \left(\frac{\mu}{k} \right) \Delta t + o(\Delta t) \right\} \quad (54)$$

$$\frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = -\lambda(t)P_0(t) + \frac{o(\Delta t)}{\Delta t} + \left(\frac{\mu}{k} \right) P_1(t) \quad (55)$$

Taking the mathematical limit on both sides as $\Delta t \rightarrow 0$, the remainder term vanishes since $\lim_{\Delta t \rightarrow 0} \frac{o(\Delta t)}{\Delta t} = 0$, yielding the initial state continuous differential equation:

$$P'_0(t) = -\lambda(t)P_0(t) + \left(\frac{\mu}{k} \right) P_1(t) \quad (56)$$

At the specific operational peak t_{max} where traffic intensity reaches λ_{max} , the derivative of the empty state probability stabilizes to zero ($P'_0(t_{max}) = 0$):

$$P'_0(t_{max}) = -\lambda(t_{max})P_0(t_{max}) + \left(\frac{\mu}{k} \right) P_1(t_{max}) = 0 \quad (57)$$

then,

$$\lambda_{max}P_0(t_{max}) = \left(\frac{\mu}{k} \right) P_1(t_{max}) \quad (58)$$

Alternatively, $\bar{\lambda}P_0 = \left(\frac{\mu}{k}\right) P_1$.

13.2 General State Probability Derivation ($n > 0$) :

For a general system state n , there are two cases:

- **Case A: For System Size within service channel capacity** ($1 \leq n < c$):

$$\begin{aligned} P_n(t + \Delta t) = & P_n(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{1 - n \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n+1}(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{(n+1) \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n-1}(t) \{\lambda(t)\Delta t + o(\Delta t)\} \left\{1 - (n-1) \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \end{aligned} \quad (59)$$

- **Case B: For System Size exceeding service channel capacity** ($n \geq c$):

$$\begin{aligned} P_n(t + \Delta t) = & P_n(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{1 - c \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n+1}(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{c \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n-1}(t) \{\lambda(t)\Delta t + o(\Delta t)\} \left\{1 - c^* \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \end{aligned} \quad (60)$$

where $c^* = c$ if $n > c$, and $c^* = c - 1$ if $n = c$.

13.3 Steady-State Solutions for Busy and Slack Hours :

By expanding these differential equations to their steady-state limits, we evaluate the system state distributions ($\bar{P}_n$ for peak busy hours, and $\underline{P}_n$ for quiet slack hours): We analyze under two steady states; one at $t_{\max}$ when λ attains $\lambda_{\max}$ & other at $t_{\min}$ when λ attains $\lambda_{\min}$

Hence, further, for the at $t = t_{\max}$ we can use *upperbar* symbol & $t = t_{\min}$ we can a *lowerbar* symbol.

13.3.1 Busy Hours State Distributions ($\bar{P}_n$):

$$\bar{P}_n = \frac{\bar{\lambda}^n}{n! \left(\frac{\mu}{k}\right)^n} \bar{P}_0, \quad 1 \leq n < c \quad (61)$$

$$\bar{P}_n = \frac{\bar{\lambda}^n}{c^{n-c} \left(\frac{\mu}{k}\right)^n c!} \bar{P}_0, \quad n \geq c \quad (62)$$

To isolate the normalization factor $\bar{P}_0$, we calculate the total probability axiom $\sum_{n=0}^{\infty} \bar{P}_n = 1$:

$$\sum_{n=0}^{c-1} \bar{P}_n + \sum_{n=c}^{\infty} \bar{P}_n = 1 \quad (63)$$

Let $\bar{\rho}$ denote the traffic intensity ratio tracking parameter during peak operating hours, defined as $\bar{\rho} = \frac{\bar{\lambda}}{c(\frac{\mu}{k})} = \frac{k\bar{\lambda}}{c\mu}$,

$$\bar{P}_0 \left[\sum_{n=0}^{c-1} \frac{(c\bar{\rho})^n}{n!} + \frac{c^c}{c!} \sum_{n=c}^{\infty} \bar{\rho}^n \right] = 1 \quad (64)$$

$$\bar{P}_0 = \frac{1}{\sum_{n=0}^{c-1} \frac{(c\bar{\rho})^n}{n!} + \frac{c^c}{c!} \{\bar{\rho}^c + \bar{\rho}^{c+1} + \dots\}} \quad (65)$$

provided that $\bar{\rho} < 1$,

$$\bar{P}_0 = \frac{1}{\sum_{n=0}^{c-1} \frac{(c\bar{\rho})^n}{n!} + \frac{c^c}{c!} \frac{\bar{\rho}^c}{1-\bar{\rho}}} \quad (66)$$

For $c = 2$,

$$\bar{P}_0 = \frac{1}{1 + c\bar{\rho} + \frac{2^2}{2!} \frac{\bar{\rho}^2}{1-\bar{\rho}}} = \frac{1}{1 + 2\bar{\rho} + \frac{2\bar{\rho}^2}{1-\bar{\rho}}} \quad (67)$$

$$\bar{P}_0 = \frac{1}{\frac{(1-\bar{\rho}) + 2\bar{\rho}(1-\bar{\rho}) + 2\bar{\rho}^2}{1-\bar{\rho}}} = \frac{1-\bar{\rho}}{1-\bar{\rho} + 2\bar{\rho} - 2\bar{\rho}^2 + 2\bar{\rho}^2} = \frac{1-\bar{\rho}}{1+\bar{\rho}} \quad (68)$$

13.3.2 Slack Hours State Distributions (P_n):

Similarly, during off-peak slack hours governed by the traffic intensity parameter $\underline{\rho} = \frac{\lambda}{c(\frac{\mu}{k})} = \frac{k\lambda}{c\mu}$ under identical stability conditions $\underline{\rho} < 1$, the empty system probability is expressed as:

$$P_0 = \frac{1}{\sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{c^c}{c!} \frac{(\rho)^c}{1-\underline{\rho}}} \quad (69)$$

For $c = 2$,

$$P_0 = \frac{1-\underline{\rho}}{1+\underline{\rho}} \quad (70)$$

14 Derivation of Mean Queue Lengths ($E(M)$) :

Let M denote the number of person in the queue when a new patient arrives at the hospital . Let N denote the number of persons in the hospital OPD. Then,

$$M = \begin{cases} 0 & \text{if } N \leq c \\ N - c & \text{if } N > c \end{cases} \quad (71)$$

To determine the mean queue length across all c operational channels, $E(M) = \sum_{n=c}^{\infty} (n - c)P_n$.

14.0.1 Expected Queue Length during Busy Hours ($E(\bar{M})$):

$$E(\bar{M}) = \sum_{n=c}^{\infty} (n - c)\bar{P}_n = \sum_{n=c}^{\infty} (n - c) \frac{\bar{\rho}^n}{c!} c^c \bar{P}_0 \quad (72)$$

$$E(\bar{M}) = \frac{c^c}{c!} \bar{P}_0 \sum_{n=c}^{\infty} (n - c) \bar{\rho}^n = \frac{c^c \bar{P}_0 \bar{\rho}^{c+1}}{c!} \sum_{n=c}^{\infty} (n - c) \bar{\rho}^{n-c-1} \quad (73)$$

Shifting $u = n - c$:

$$E(\bar{M}) = \frac{c^c \bar{P}_0 \bar{\rho}^{c+1}}{c!} \sum_{u=0}^{\infty} u \bar{\rho}^{u-1} = \frac{c^c \bar{P}_0}{c!} [1 + 2\bar{\rho} + 3\bar{\rho}^2 + \dots] \bar{\rho}^{c+1} \quad (74)$$

Using a geometric series $\sum_{u=0}^{\infty} u \bar{\rho}^{u-1} = \frac{1}{(1-\bar{\rho})^2}$:

$$E(\bar{M}) = \frac{c^c \bar{P}_0}{c!} \frac{1}{(1 - \bar{\rho})^2} (\bar{\rho})^{c+1} \quad (75)$$

For $c = 2$,

$$E(\bar{M}) = \frac{2^2 \bar{P}_0}{2!} \frac{\bar{\rho}^{2+1}}{(1 - \bar{\rho})^2} = 2\bar{P}_0 \frac{\bar{\rho}^3}{(1 - \bar{\rho})^2} \quad (76)$$

14.0.2 Expected Queue Length during Slack Hours ($E(\underline{M})$):

During off-peak slots for $c = 2$ is expressed as:

$$E(\underline{M}) = 2P_0 \frac{(\underline{\rho})^3}{(1 - \underline{\rho})^2} \quad (77)$$

15 Derivation of Mean Number of Persons in the System ($E(N)$) :

The total expected number of patients residing within the absolute system is given by $E(N) = \sum_{n=0}^{\infty} nP_n$.

15.0.1 Mean System Count during Busy Hours ($E(\bar{N})$):

$$E(\bar{N}) = \sum_{n=0}^{\infty} n\bar{P}_n = \sum_{n=0}^{c-1} n\bar{P}_n + \sum_{n=c}^{\infty} n\bar{P}_n \quad (78)$$

$$E(\bar{N}) = \sum_{n=0}^{c-1} n \frac{(c\bar{\rho})^n}{n!} \bar{P}_0 + \sum_{n=c}^{\infty} n \frac{c^c (\bar{\rho})^n \bar{P}_0}{c!} \quad (79)$$

Splitting the coordinate index within the upper summation boundary by rewriting $n = (n - c) + c$:

$$E(\bar{N}) = \sum_{n=1}^{c-1} \frac{n(c\bar{\rho})^n}{n!} \bar{P}_0 + \frac{c^c \bar{P}_0}{c!} \sum_{n=c}^{\infty} ((n - c) + c) \bar{\rho}^n \quad (80)$$

$$E(\bar{N}) = \sum_{n=1}^{c-1} \frac{(c\bar{\rho})^n}{(n-1)!} \bar{P}_0 + \frac{c^c \bar{P}_0}{c!} \left\{ \bar{\rho}^{c+1} \sum_{u=0}^{\infty} u \bar{\rho}^{u-1} + c \bar{\rho}^c \sum_{u=0}^{\infty} \bar{\rho}^u \right\} \quad (81)$$

$$E(\bar{N}) = \sum_{n=1}^{c-1} \frac{(c\bar{\rho})^n}{(n-1)!} \bar{P}_0 + \frac{c^c \bar{P}_0}{c!} \left\{ \bar{\rho}^{c+1} \frac{1}{(1 - \bar{\rho})^2} + \frac{c \bar{\rho}^c}{(1 - \bar{\rho})} \right\} \quad (82)$$

15.0.2 Mean System Count during Slack Hours ($E(\underline{N})$) :

The mathematical expression tracking the absolute patient metrics within the total facility during off-peak periods, denoted as $E(\underline{N})$, is obtained directly by substituting the parameters $\bar{\rho} \rightarrow \underline{\rho}$ and $\bar{P}_0 \rightarrow \underline{P}_0$ in the structural formula above.

16 Derivation of Waiting Time in the Queue (T_q):

The probability that an arriving patient has to wait before purchasing a ticket corresponds to the condition that all c registration operators are occupied, requiring at least c patients to be in the system:

$$P[\text{Arriving patient must wait}] = \sum_{n=c}^{\infty} P_n \quad (83)$$

This probability evaluates to $\sum_{n=c}^{\infty} \bar{P}_n$ during busy blocks and $\sum_{n=c}^{\infty} P_n$ during slack hours.

Let T_q denote the waiting time in the queue of an arriving patients. For a non-zero waiting duration ($t > 0$),

$$F_{T_q}(t) = P[T_q \leq t] = P[T_q = 0] + P[0 < T_q \leq t] \quad (84)$$

Conditioning across the system states $N = n$ when the new arrival joins the line:

$$F_{T_q}(t) = \sum_{n=0}^{c-1} P_n + \sum_{n=c}^{\infty} P_n P[0 < T_q \leq t \mid N = n] \quad (85)$$

When $n \geq c$, an arriving patient must wait for the completion of $(n - c + 1)$ distinct phases across the c parallel servers. :

$$F_{T_q}(t) = \sum_{n=0}^{c-1} P_n + \sum_{n=c}^{\infty} P_n \int_0^t \frac{\left(c \left(\frac{\mu}{k}\right)\right)^n}{\Gamma(n)} e^{-c\left(\frac{\mu}{k}\right)z} z^{n-1} dz \quad (86)$$

$$F_{T_q}(t) = \sum_{n=0}^{c-1} P_n + \sum_{n=c}^{\infty} \frac{c^c}{c!} \rho^n P_0 \int_0^t \frac{\left(c \left(\frac{\mu}{k}\right)\right)^n}{\Gamma(n)} e^{-c\left(\frac{\mu}{k}\right)z} z^{n-1} dz \quad (87)$$

This expression involves an incomplete Gamma integral up to $c \left(\frac{\mu}{k}\right) t$ with shape parameter n . To extract the probability density function (p.d.f.) $f_{T_q}(t)$ for $t > 0$, we differentiate $F_{T_q}(t)$ with respect to t using the Leibniz integral rule:

Let $\omega = c \left(\frac{\mu}{k}\right) z$, which gives $d\omega = c \left(\frac{\mu}{k}\right) dz$. The internal integral transformation simplifies as:

$$\int_0^t \frac{\left(c \left(\frac{\mu}{k}\right)\right)^n}{\Gamma(n)} e^{-c\left(\frac{\mu}{k}\right)z} z^{n-1} dz = \int_0^{c\left(\frac{\mu}{k}\right)t} \frac{e^{-\omega} \omega^{n-1}}{\Gamma(n)} d\omega \quad (88)$$

Differentiating this transformed bound with respect to t :

$$f_{T_q}(t) = \sum_{n=c}^{\infty} \frac{c^c}{c!} \rho^n \frac{d \left(c \left(\frac{\mu}{k}\right) t\right)}{dt} \cdot \frac{d}{d \left(c \left(\frac{\mu}{k}\right) t\right)} \int_0^{c\left(\frac{\mu}{k}\right)t} \frac{e^{-\omega} \omega^{n-1}}{\Gamma(n)} d\omega \quad (89)$$

$$f_{T_q}(t) = \sum_{n=c}^{\infty} \frac{c^c}{c!} \rho^n \frac{c \left(\frac{\mu}{k}\right)}{\Gamma(n)} e^{-c\left(\frac{\mu}{k}\right)t} \left(c \left(\frac{\mu}{k}\right) t\right)^{n-1} \quad (90)$$

$$f_{T_q}(t) = e^{-c\left(\frac{\mu}{k}\right)t} \frac{c^c}{c!} \rho \left(\frac{\mu}{k}\right) \sum_{n=c}^{\infty} \frac{\left(c \left(\frac{\mu}{k}\right) t \rho\right)^{n-1}}{(n-1)!} \quad (91)$$

$$f_{T_q}(t) = e^{-c\left(\frac{\mu}{k}\right)t} \frac{c^c}{c!} \rho \left(\frac{\mu}{k}\right) \left[e^{c\left(\frac{\mu}{k}\right)t\rho} - \sum_{n=0}^{c-2} \frac{\left(c \left(\frac{\mu}{k}\right) t \rho\right)^n}{n!} \right] \quad (92)$$

16.0.1 Derivation of Mean Waiting Time in the Queue ($E(T_q)$):

The expected waiting time of an arriving patient in the queue is found by integrating the definition $E(T_q) = \int_0^\infty t \cdot f_{T_q}(t) dt$:

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\int_0^\infty t \cdot e^{-(1-\rho)c \left(\frac{\mu}{k} \right) t} dt - \sum_{n=0}^{c-2} \frac{1}{n!} \int_0^\infty t \cdot e^{-c \left(\frac{\mu}{k} \right) t} \left(c \left(\frac{\mu}{k} \right) t \rho \right)^n dt \right] \quad (93)$$

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \sum_{n=0}^{c-2} \frac{1}{n!} \left(c \left(\frac{\mu}{k} \right) \rho \right)^n \frac{(n+1)!}{\left(c \left(\frac{\mu}{k} \right) \right)^{n+2}} \right] \quad (94)$$

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \sum_{n=0}^{c-2} \frac{\rho^n (n+1)}{c^2 \left(\frac{\mu}{k} \right)^2} \right] \quad (95)$$

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{k} \right)^2} \left\{ \rho \sum_{n=1}^{c-2} n \rho^{n-1} + \sum_{n=0}^{c-2} \rho^n \right\} \right] \quad (96)$$

Let $S_c = 1 + 2\rho + 3\rho^2 + \dots + (c-2)\rho^{c-3}$. Multiplying through by the scaling factor ρ yields $\rho S_c = \rho + 2\rho^2 + \dots + (c-2)\rho^{c-2}$. Subtracting these expressions forms a clean telescoping sequence:

$$(1-\rho)S_c = 1 + \rho + \rho^2 + \dots + \rho^{c-3} - (c-2)\rho^{c-2} = \frac{1 - \rho^{c-2}}{1 - \rho} - (c-2)\rho^{c-2} \quad (97)$$

$$S_c = \frac{1 - \rho^{c-2}}{(1-\rho)^2} - \frac{(c-2)\rho^{c-2}}{1-\rho} \quad (98)$$

Substituting this summation series back into the expression establishes the explicit calculation layout:

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{k} \right)^2} \left\{ \frac{1 - \rho^{c-1}}{1 - \rho} + \frac{\rho(1 - \rho^{c-2})}{(1-\rho)^2} - \frac{(c-2)\rho^{c-1}}{1 - \rho} \right\} \right] \quad (99)$$

17 Total System Waiting Time Derivation ($E(W)$):

The total mean waiting time of an arriving patient across the entire hospital OPD registration system is defined via conservation equations as:

$$E(W) = E(T_q) + \text{Mean Service Time} \quad (100)$$

$$E(W) = E(T_q) + E\left(\frac{1}{\mu_n}\right) \quad (101)$$

Case-1: Approximate the second term as,

$$E\left(\frac{1}{\mu_n}\right) \approx \frac{k}{c\mu} \quad (102)$$

Then,

$$E(W) = E(T_q) + \frac{k}{c\mu} \quad (103)$$

Case-2:

Because the system configuration contains a state-dependent service mechanism , the expected service rate $E\left(\frac{1}{\mu_n}\right)$ must be averaged across all valid probability coordinates P_n :

$$E\left(\frac{1}{\mu_n}\right) = \sum_{n=1}^{\infty} \frac{1}{\mu_n} P_n \quad (104)$$

Now we can derived it as,

$$\begin{aligned}
E\left(\frac{1}{\mu_n}\right) &= \frac{K}{\mu} \sum_{n=1}^{c-1} \frac{1}{n} P_n + \frac{K}{c\mu} \sum_{n=c}^{\infty} P_n \\
&= \frac{K}{\mu} \sum_{n=1}^{c-1} \frac{P_n}{n} + \frac{K}{c\mu} \left(1 - \sum_{n=0}^{c-1} P_n\right) \\
&= \frac{K}{\mu} \frac{\lambda^n}{n! \left(\frac{\mu}{K}\right)^n} \cdot P_0 + \frac{K}{c\mu} \left(1 - \sum_{n=0}^{c-1} \frac{\lambda^n}{n! \left(\frac{\mu}{K}\right)^n} P_0\right) \\
&= \frac{K^2}{\mu^2} \frac{\lambda^n}{n!} \cdot P_0 + \frac{K}{c\mu} \left(1 - \sum_{n=0}^{c-1} \frac{\lambda^n}{n!} \left(\frac{K}{\mu}\right)^n P_0\right) \\
&= \frac{K^2}{\mu^2} \lambda P_0 + \frac{K}{c\mu} \left(1 - P_0 - \lambda \frac{K}{\mu} P_0\right) \\
&= \frac{\lambda K^2}{\mu^2} P_0 \left(1 - \frac{1}{c}\right) + \frac{K}{c\mu} (1 - P_0)
\end{aligned}$$

$$\begin{aligned}
\text{Now, } E\left(\frac{1}{\mu_n}\right) &= \lambda \cdot \left(\frac{K}{\mu}\right)^2 \cdot P_0 \left(1 - \frac{1}{c}\right) + \frac{K}{c\mu} (1 - P_0) \\
&= \frac{\lambda K^2}{\mu^2} \frac{1 - \rho}{1 + \rho} \left(1 - \frac{1}{c}\right) + \frac{K}{c\mu} \left(1 - \left(\frac{1 - \rho}{1 + \rho}\right)\right)
\end{aligned}$$

Case-3:

By Little's Formula we can write it as,

$$E(W) = \frac{E(N)}{\lambda} \quad (105)$$

18 Numerical Output of Our Queueing Model

Given parameters and bounds:

$$\mu/K = 1.42, \quad \bar{\lambda} = 1.5, \quad \underline{\lambda} = 0.2, \quad c = 2$$

The traffic intensity for busy and slack hours respectively,

$$\bar{\rho} = \frac{\bar{\lambda} \cdot K}{2 \cdot \mu} = 0.58, \quad \underline{\rho} = \frac{\underline{\lambda} \cdot K}{2 \cdot \mu} = 0.07$$

During busy hours, the traffic intensity reaches $\bar{\rho} = 0.58$, meaning the servers are

moderately active but stable. During slack hours, it drops to a very low $\underline{\rho} = 0.07$.

For $c = 2$:

$$\bar{P}_0 = \frac{1 - \bar{\rho}}{1 + \bar{\rho}} = \frac{1 - 0.58}{1 + 0.58} = \frac{0.42}{1.58} = 0.2658 \approx 0.27$$

$$\underline{P}_0 = \frac{1 - \underline{\rho}}{1 + \underline{\rho}} = \frac{1 - 0.07}{1 + 0.07} = \frac{0.93}{1.07} \approx 0.87$$

$E(M)$ = Mean Queue Length Combining All Queues

For busy hours,

$$E(\bar{M}) = \frac{2 \cdot \bar{P}_0 \cdot \bar{\rho}^3}{(1 - \bar{\rho})^2} = \frac{2 \times 0.27 \times (0.58)^3}{(1 - 0.58)^2} = \frac{0.10536}{0.1764} = 0.5972 \approx 0.597$$

For slack hours,

$$E(\underline{M}) = \frac{2 \cdot \underline{P}_0 \cdot \underline{\rho}^3}{(1 - \underline{\rho})^2} = \frac{2 \times 0.87 \times (0.07)^3}{(1 - 0.07)^2} = \frac{0.008526}{0.8649} = 0.0098$$

On average, during peak times, there is about 0.60 of a person waiting in line. On the other hand, during off-peak, the average number of people waiting drops to 0.01.

$E(N)$ = mean numbers of persons in the hospital OPD

$$E(N) = \sum_{n=1}^{c-1} \frac{(c\rho)^n P_0}{n!} + \frac{c^c P_0}{c!} \left\{ \frac{\rho^{c+1}}{(1 - \rho)^2} + \frac{c\rho^c}{1 - \rho} \right\}$$

In our case $c = 2$, so $n = 1$:

For busy hours,

$$\begin{aligned} E(\bar{N}) &= \frac{(2\bar{\rho})^1 \bar{P}_0}{1!} + \frac{2^2 \bar{P}_0}{2!} \left\{ \frac{\bar{\rho}^3}{(1 - \bar{\rho})^2} + \frac{2\bar{\rho}^2}{1 - \bar{\rho}} \right\} \\ &= \frac{2 \times 0.58 \times 0.27}{1} + \frac{4 \times 0.27}{2} \left\{ \frac{(0.58)^3}{(0.42)^2} + \frac{2 \times (0.58)^2}{0.42} \right\} \\ &= 0.3132 + \frac{1.08}{2} \times \left\{ \frac{0.195112}{0.1764} + \frac{0.6728}{0.42} \right\} \\ &= 0.3132 + 0.54 \times (1.10607 + 1.60190) \\ &= 0.3132 + 0.54 \times 2.70797 \\ &= 0.3132 + 1.4623 = 1.7755 \end{aligned}$$

For slack hours,

$$\begin{aligned}
E(\underline{N}) &= \frac{2 \times 0.07}{1} \times 0.87 + \frac{2^2 \times 0.87}{2!} \left\{ \frac{(0.07)^3}{(1 - 0.07)^2} + \frac{2 \times (0.07)^2}{1 - 0.07} \right\} \\
&= 0.1218 + 1.74 \left(\frac{0.000343}{0.8649} + \frac{0.0098}{0.93} \right) \\
&= 0.1218 + 1.74(0.0003966 + 0.010537) \\
&= 0.1218 + (1.74 \times 0.0109336) \\
&= 0.1218 + 0.01902 \\
&= 0.14082
\end{aligned}$$

During peak hours, there are typically 1.78 patients in the OPD system at any given moment. On the other hand, during off-peak hours, the average number of patients drops significantly to just 0.14.

$E(T_q)$ = Expected Waiting Time in Queue

$$E(\overline{T}_q) = \frac{c^c}{c!} \bar{\rho}^c \left(\frac{\mu}{K} \right) \left[\frac{1}{(1 - \bar{\rho})^2 c^2 \left(\frac{\mu}{K} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{K} \right)^2} \right] \left\{ \frac{1 - \bar{\rho}^{c-1}}{1 - \bar{\rho}} + \frac{\bar{\rho}(1 - \bar{\rho}^{c-2})}{(1 - \bar{\rho})^2} - \frac{(c - 2)\bar{\rho}^{c-1}}{1 - \bar{\rho}} \right\}$$

Evaluating for $c = 2$:

For busy hours,

$$\begin{aligned}
E(\overline{T}_q) &= \frac{2^2 \times 0.58 \times 2 \times 1.42}{2!} \left[\frac{1}{(1 - 0.58)^2 \times 4 \times (1.42)^2} - \frac{1}{4 \times (1.42)^2} \right] \left[\frac{1 - 0.58}{1 - 0.58} + \frac{0.58(1 - 0.58)^0}{(1 - 0.58)^2} \right] \\
&= \frac{6.5888}{2} \left[\frac{1}{0.1764 \times 2.0164} - \frac{1}{8.0656} \right] [1 + 3.2879 - 0] \\
&= 3.2944 \left(\frac{1}{0.35569} - \frac{1}{8.0656} \right) \\
&= 3.2944(2.8114 - 0.1239) \\
&= 3.2944 \times 0.5789 \\
&= 1.9071
\end{aligned}$$

For slack hours,

$$\begin{aligned}
E(\underline{T}_q) &= \frac{4 \times 0.07 \times 2 \times 1.42}{2!} \left[\frac{1}{(1 - 0.07)^2 \cdot 4(1.42)^2} - \frac{1}{4 \times (1.42)^2} \right] \left\{ \frac{1 - 0.07}{1 - 0.07} + \frac{\rho(1 - (0.07)^0)}{(1 - 0.07)^2} - 0 \right\} \\
&= 0.3976 \left(\frac{1}{7.501008 \times 8.0656} - \frac{1}{8.0656} \right) (1 + 0 - 0) \\
&= 0.3976 \times (0.13331 - 0.12398) \\
&= 0.0037
\end{aligned}$$

During busy hours, patients wait about 2 minutes on average before purchasing a ticket. During slack hours, the waiting time is almost negligible, meaning patients are served nearly immediately.

Now, we are estimating total system waiting time by 3 ways,

Case-1:

For busy hours,

$$E(\overline{W}) = E(\overline{T}_q) + \frac{k}{c\mu} = 1.9071 + 0.351 = 2.258 \quad (106)$$

For Slack hours,

$$E(\underline{W}) = E(\underline{T}_q) + \frac{k}{c\mu} = 0.0037 + 0.351 = 0.351 \quad (107)$$

On average, a patient spends about 2.26 minutes in the OPD system during busy hours and only 0.35 minutes during slack hours. This shows that the ticketing system is efficient and patients are served quickly throughout the day.

Case-2:

Now to calculate,

$$\begin{aligned} E\left(\frac{1}{\underline{\mu}_n}\right) &= \left(\frac{1}{1.42}\right)^2 \times 1.5 \times 0.27 \times 0.5 + \frac{1}{1.42 \times 2}(1 - 0.27) \\ &= 0.101 + 0.257 \\ &= 0.358 \end{aligned}$$

$$\begin{aligned} E\left(\frac{1}{\underline{\mu}_n}\right) &= \left(\frac{1}{1.42}\right)^2 \times 0.2 \times 0.87 \times 0.5 + \frac{1 - 0.87}{1.42 \times 2} \\ &= 0.586 + 0.0457 \\ &= 0.632 \end{aligned}$$

For busy hours,

$$E(\overline{W}) = 1.9071 + 0.358 = 2.26$$

For slack hours,

$$E(\underline{W}) = 0.0037 + 0.632 = 0.635$$

During busy hours, a patient spends about 2.26 minutes in the OPD system. During slack hours, the time reduces to only 0.63 minutes.

Case-3: Now by Little's formula

$$E(W) = \frac{E(N)}{\lambda}$$

For busy hours,

$$E(\overline{W}) = \frac{1.775}{1.5} = 1.133$$

For slack hours,

$$E(\underline{W}) = \frac{0.1408}{0.2} = 0.70$$

Using Little's Formula, the average time spent by a patient in the OPD system is about 1.13 minutes during busy hours and 0.70 minutes during slack hours.

19 Conclusion :

In this project, an attempt has been made to study the queueing system of the OPD ticket counter at Barasat Government Medical College and Hospital using real observational data. From the analysis, it was found that the arrival of patients changes with time and does not remain constant throughout the day. Because of this time-varying nature, the arrival process was modeled by a Non-Homogeneous Poisson Process (NHPP) instead of the usual homogeneous Poisson process. The estimated arrival rate function was able to describe the increase and decrease of patient flow during different time intervals, especially during busy hours.

The study of inter-arrival times showed approximately exponential behaviour for smaller intervals, which supports the use of Poisson-type arrivals. On the other hand, the service time data did not fit the exponential distribution properly. Different Erlang models were compared, and it was observed that Erlang distributions provide a better description of the service mechanism. Although Erlang-6 gave the minimum AIC value, Erlang-3 was finally selected since it gives a reasonable balance between goodness of fit and mathematical simplicity. Thus, the service process can be viewed as a 3-Erlang Renewal Count Process.

Based on these observations, a generalized queueing model with time-dependent arrivals and Erlang-type service was formulated. Various system characteristics such as state probabilities, queue length, waiting behaviour, and congestion during peak and non-peak periods were studied mathematically. The results indicate that practical hospital queueing systems are more complex than the standard M/M/c assumptions and therefore require more realistic stochastic models.

The overall study shows that queueing theory can be effectively applied to healthcare systems for understanding patient flow and service efficiency. The findings of this work may help hospital authorities in better management of OPD counters, reduction of waiting time, and improvement of service quality. The methodology used in this project may also be useful for analyzing other real-life service systems where arrivals vary with time and services occur in multiple stages.

The results indicate that the existing two-counter registration system is sufficient for the current patient load. Since the waiting time and queue length are relatively low, there is no immediate need for an additional counter or faster service under the present conditions.

20 References :

References

- [1] J. Medhi, *Stochastic Models in Queueing Theory*, 2nd ed. San Diego, CA, USA: Academic Press, 2002.
- [2] D. Gross, J. F. Shortle, J. M. Thompson, and C. M. Harris, *Fundamentals of Queueing Theory*, 4th ed. Hoboken, NJ, USA: John Wiley & Sons, 2008.
- [3] Sheldon M. Ross, *Stochastic Process*, 2nd ed., USA: John Wiley & Sons, 1996.

21 Appendix :

The statistical analysis and graphical visualizations presented in this project were carried out using R software. The corresponding R programs were used for data preprocessing, distribution fitting, parameter estimation, goodness-of-fit testing, and numerical computation of the queueing model characteristics.

To ensure reproducibility and transparency, all relevant R scripts, datasets, and supporting outputs have been organized and made available in a GitHub repository. Interested readers may refer to the repository for implementation details and additional computational results.

All R scripts and supporting materials are available in the following GitHub repository:
<https://github.com/STAT-1092/STAT1092-Project-Research-2026>